

EUNJEONG PARK

Ph.D. Student in Mechanical Engineering, The Pennsylvania State University
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RESEARCH INTERESTS

My research focuses on machine learning and physics-based modeling for materials and manufacturing systems. I am particularly interested in physics-informed machine learning, multiphysics simulation, inverse design, and data-driven approaches for predicting and optimizing compositional, microstructural, and structural behavior in advanced materials and additive manufacturing.

EDUCATION

The Pennsylvania State University, United States

Aug. 2025 – May 2029

- Ph.D. in Mechanical Engineering, GPA: 3.9/4.0
- Advisor: Prof. Amrita Basak
- Thesis: AI-enabled Compositional and Microstructure Evaluation

Korea Advanced Institute of Science and Technology (KAIST), South Korea *Sep. 2020 – Aug. 2022*

- M.S. in Mechanical Engineering, GPA: 3.9/4.3
- Advisor: Prof. Seunghwa Ryu
- Thesis: Machine Learning-based Elastic Modulus Prediction and Structural Shape Optimization

Inha University, South Korea

Feb. 2017 – Aug. 2020

- B.S. in Mechanical Engineering, GPA: 4.3/4.5 (Early Graduation)

EXPERIENCE

The Pennsylvania State University

Aug. 2025 – Present

Graduate Research Assistant, Mechanical Engineering

- Developing physics-informed machine learning frameworks, including graph neural networks and large language models, for multicomponent alloy phase prediction, out-of-distribution generalization, and inverse alloy design.
- Developing physics-guided generative AI models, including diffusion models, for process-conditioned microstructure image generation in laser directed energy deposition (L-DED).

Samsung Electronics Semiconductor R&D Center

Aug. 2022 – Jul. 2025

R&D Engineer

- Developed a wafer-level inspection test bench targeting full-wafer inspection, including optical feasibility evaluation and experimental system development.
- Conducted acoustic and photoacoustic simulations using COMSOL Multiphysics and investigated photoacoustic techniques for nondestructive detection of bonding defects in semiconductor structures.
- Supported Joint Development and Execution Plan (JDEP) projects for semiconductor equipment development, including technical feasibility assessment, experimental evaluation, and collaboration with external equipment partners.
- Developed and operated process recipes for semiconductor equipment, including process condition setup, parameter adjustment, and performance evaluation.

Korea Advanced Institute of Science and Technology (KAIST)*Aug. 2020 – Aug. 2022**Graduate Research Assistant*

- Developed structural mechanics simulations using COMSOL Multiphysics to evaluate the effective mechanical properties of patterned structures.
- Generated simulation datasets and developed Multi-objective Gaussian Process models for prediction of effective elastic properties.
- Performed multi-objective shape optimization to identify structural geometries with improved mechanical performance.

Inha University – 3D Printing & Laser Cutting Center*Mar. 2020 – Jul. 2020**Student Assistant*

- Assisted in the operation and maintenance of 3D printers and laser cutting equipment.
- Supported students and faculty with equipment operation, fabrication, and safety procedures.

KIA Motors*Sep. 2019 – Dec. 2019**Intern*

- Contributed to an electric vehicle commercialization project focused on the development and implementation of a battery swapping system.

HONORS & AWARDS

Aug.2025	University Graduate Fellowship , <i>The Pennsylvania State University</i> – Awarded to outstanding incoming Ph.D. students in recognition of academic excellence and research potential.
Mar.2020	Hanjin Foundation Scholarship , <i>Hanjin Group</i> – Selected as a recipient of a competitive scholarship supporting high-achieving undergraduate students.
Feb.2020	Super Challenge Hackathon, Best Award , <i>Inha University Start-up Center</i> .
Feb.2019	University Simulation Competition, Honorable Mention , <i>MSC Software</i> .
Oct.2018	3D Printing Creative Competition, Grand Prize , <i>KSPSE</i> .
Sep.2018	Comprehensive Design Competition, Grand Prize , <i>Inha University Innovation Center</i> .

JOURNAL PAPERS

- [2] **Eunjeong Park** and Amrita Basak. “Multi-Label Phase Diagram Prediction in Complex Alloys via Physics-Informed Graph Attention Networks.” *arXiv preprint arXiv:2604.16468* (2026).
- [1] Hyunggi Song, **Eunjeong Park**, Hong Jae Kim, Chung-Il Park, Taek-Soo Kim, Yoon Young Kim, and Seunghwa Ryu. “Free-form optimization of pattern shape for improving mechanical characteristics of a concentric tube.” *Materials & Design* 230 (2023): 111974.

CONFERENCE PRESENTATIONS

- [4] **Eunjeong Park** and Amrita Basak. “Out-of-Distribution Phase Equilibrium Prediction Using Physics-Informed Graph Attention Networks.” *2027 TMS Annual Meeting & Exhibition, AI/ML/Data Informatics for Materials Discovery Symposium, Accepted for Oral Presentation*, 2027.
- [3] **Eunjeong Park** and Amrita Basak. “Physics-Informed Graph Attention Networks for Accurate Multi-Phase Prediction in Ag–Bi–Cu–Sn Alloys.” *College of Engineering Research Symposium (CERS), The Pennsylvania State University*, 2026.

- [2] **Eunjeong Park** and Seunghwa Ryu. “Machine Learning-based Pattern Shape Optimization to Improve the Mechanical Performance of Concentric Tube.” *The Korean Society of Mechanical Engineers Annual Conference*, 2022.
- [1] Jiyoung Jeong, Yongtae Kim, **Eunjeong Park**, and Seunghwa Ryu. “Transfer Learning Framework Utilizing Homogenization Theory to Predict the Effective Behavior of Elasto-Plastic Composites.” *The Korean Society of Mechanical Engineers Annual Conference*, 2021.

RESEARCH MENTORING

High School Student Research Mentoring: Juliet Rueda

Mar. 2026 – Present

- Mentoring an 11th-grade student on a machine-learning framework for predicting material properties of Al_2O_3 –Cu composites.
- Guiding feature design using phase maps and grain-level microstructural descriptors for structure–property prediction.
- Advising on research framing and scientific communication; the project received third place in the Somers High School Science Research Fair poster category.

REVIEWING ACTIVITIES

- [1] **Peer Reviewer**, *Engineering Applications of Artificial Intelligence (Elsevier)*, 2026.

ACADEMIC SKILLS

Programming Languages	Python, MATLAB
AI & Machine Learning	Machine Learning, Deep Learning, Gaussian Process, Large Language Models, Vision Transformer (ViT), Diffusion Models
Simulation & Analysis	COMSOL Multiphysics, ANSYS, ADAMS, Monte Carlo, CALPHAD
Fabrication & Optics	Metal 3D Printing, Polymer 3D Printing, Microscopy, AFM, SAM, e-Beam, WLI
CAD	Solid Edge, NX, AutoCAD

LANGUAGES

Korean (Native) **English** (Working Proficiency)